Separation and Purification Technology 289 (2022) 120736

Contents lists available at ScienceDirect

Separation and Purification Technology

journal homepage: www.elsevier.com/locate/seppur

Efficient separation of CO_2/CH_4 by ionic liquids confined in graphene oxide: A molecular dynamics simulation

Fang Yan [a,b], Yandong Guo [a,*], Zhenlei Wang [b,c], Linlin Zhao [a,b], Xiaochun Zhang [b,*]

[a] College of Mathematics Science, Bohai University, Jinzhou, Liaoning 121013, China
[b] Innovation Academy for Green Manufacture, Beijing Key Laboratory of Ionic Liquids Clean Process, CAS Key Laboratory of Green Process and Engineering, State Key Laboratory of Multiphase Complex Systems, Institute of Process Engineering, Chinese Academy of Sciences, Beijing 100190, China
[c] College of Chemistry and Chemical Engineering, Qingdao University, Qingdao, Shandong 266071, China

ARTICLE INFO

Keywords:
Ionic liquids
Graphene oxide
CO_2/CH_4 separation
Molecular dynamics simulation

ABSTRACT

Ionic liquids (ILs)/graphene oxide (GO) membranes have been regarded as a prospective alternative in CO_2 separation. However, the correlation between the microstructure of ILs/GO and CO_2 separation performance is unclear. In this work, the dynamic properties and interactions for CO_2/CH_4 in 1-butyl-3-methylimidazolium hexafluorophosphate ([Bmim][PF_6]), 1-butyl-3-methylimidazolium tetrafluoroborate ([Bmim][BF_4]) and 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([Bmim][TF_2N]) confined between two GO sheets with different layer spacings were studied using molecular dynamics simulation. The number density and angular orientation of cations suggest that there is a dense cation adsorption layer near GO with the imidazole rings and alkyl side chains mostly paralleled to GO surface. The strong interaction of GO-cations weakens the interaction of cations-anions, facilitating the adsorption of CO_2. The overlapping distribution regions of the number density of CO_2/CH_4 with the cations and anions reveal that CO_2/CH_4 mainly distribute around the ILs area of the ILs/GO membrane, which improves the gas selectivity. The RDFs results of CO_2/CH_4-ILs indicate that the confined ILs/GO system is more favorable for capturing gases than bulk ILs. The stronger interaction of CO_2-anions/cations and the faster diffusion of CO_2 than CH_4 reflect the high solubility selectivity and diffusion selectivity for CO_2/CH_4 in ILs/GO membrane. In addition, it was found that decreasing the layer spacing would increase the solubility selectivity, but could decrease the diffusion selectivity. However, solubility selectivity plays a dominant role, thus 2 nm is the optimal layer spacing. Furthermore, the low viscosity ILs were found to be beneficial to improve the diffusion selectivity. Finally, [Bmim][TF_2N]/GO membrane is predicted to possess superior CO_2/CH_4 separation performance than [Bmim][PF_6]/GO and [Bmim][BF_4]/GO membranes.

1. Introduction

The world is facing climate and environmental problems caused by emissions of greenhouse gases, specifically CO_2[1,2]. Governments are responding to the challenge of CO_2 reduction by proposing carbon peaking and carbon neutrality plans[3,4]. The capture and separation of CO_2 are crucial to achieve these goals[5,6].

Many technologies have been investigated and utilized to capture and separate CO_2, such as absorption, adsorption, and membrane separation[7–9]. Among all these technologies, membrane separation offers a prospective option owing to its compactness, energy efficiency, and simplicity of operation[10–17]. In recent years, researchers have developed various novel two-dimensional nanomaterials with the single atomic layer thickness and high thermal stability for CO_2 separation, such as graphene-based membrane materials[18–26], boron nitride [27,28], metal-organic frameworks (MOFs)[29–33] and molybdenum disulfide (MoS_2)[34–37].

On the other hand, ionic liquids (ILs) have also attracted great interest in the field of CO_2 capture and separation due to their high CO_2 solubility, low volatility, stable performance, designable structure and environmental friendliness[38–42]. For example, Brennecke et al.[43] firstly reported the solubility of CO_2 in ILs [Bmim][PF_6], reaching a mole fraction of 0.6 at 13.8 MPa and 313.15 K in 1999. Then, they[44] found [TF_2N] anion increases CO_2 solubility relative to [BF_4] anion and [PF_6] anion based ILs. The solubilities of CO_2 in [Bmim][TF_2N], [Bmim][BF_4] and [Bmim][PF_6] were 0.17, 0.10 and 0.11 at 323.15 K and 10

* Corresponding authors.
E-mail addresses: gyd1030@163.com (Y. Guo), xchzhang@ipe.ac.cn (X. Zhang).

https://doi.org/10.1016/j.seppur.2022.120736
Received 10 January 2022; Received in revised form 15 February 2022; Accepted 23 February 2022
Available online 24 February 2022
1383-5866/

bar, respectively.

Therefore, the membrane composed of ILs and two-dimensional nanomaterials (e.g. GO) has been widely investigated for CO_2 separation recently, especially in experiments[21,45–48]. For example, Chae *et al.*[45] combined GO/copper nanoparticles (CuNP) with 1-hexyl-3-methylimidazolium nitrate ([Hmim][NO_3]) or 1-methyl-3-octylimidazolium tetrafluoroborate ([Moim][BF_4]) to prepare the facilitated CO_2 transport membrane. They found that the CO_2/N_2 separation in [Hmim][NO_3]/CuNP/GO and [Moim][BF_4]/CuNP/GO membranes were 8.7 and 30.7, respectively. The CO_2 permeability in the ILs/CuNP/GO was enhanced compared to that in the ILs/CuNP. Ying *et al.*[47] fabricated CO_2-philic membranes by confining 1-butyl-3-methylimidazolium tetrafluoroborate [Bmim][BF_4] within a layered GO membrane. The CO_2 permeability was 68.5 GPU, and the CO_2/H_2, CO_2/CH_4 and CO_2/N_2 selectivity were 24, 234 and 382, respectively, which are superior to the 2008 Robeson upper bound. After that, Ying *et al.*[46] improved the separation performance by applying an external electric field to ILs-filled GO nanoslits, doubling the CO_2 permeability and further improving the selectivity of CO_2. Furthermore, Lee *et al.*[48] immobilized poly(diallyldimethylammonium) 2-cyanopyrrolidine (P[DADMA][2-CNpyr]) and 1-ethyl-3-methylimidazolium 2-cyanopyrrolidine [Emim][2-CNpyr] within GO sheets for separating CO_2/N_2. The membrane exhibited excellent durability with a CO_2 permeability of 3090 GPU and a CO_2/N_2 selectivity of 1180. All these experimental results support that the separation membranes made by combining ILs and GO exhibit high CO_2 permeability and selectivity.

Meanwhile, the computer simulation methods such as molecular dynamics (MD) simulations have been used in a wide range for studying microscopic mechanisms in the ILs systems[49,50], and there is also some work related to the ILs/GO system[51–54]. For example, Wang *et al.*[52] adopted non-equilibrium MD simulations to analyze the structure and flow properties of 1-ethyl-3-methylimidazolium bis [(trifluoromethyl)sulfonyl] imide [Emim][TFSI] in nanochannels constructed from graphene or GO. The constraint was found to have a dominant effect on the viscosity of the ILs, but with little effect on the interfacial friction coefficient. Tang *et al.*[53] discussed the adsorption performance of GO and N-octylpyridine bis(trifluoromethyl)sulfonylimine [OPY][TF_2N]-modified GO (ILs-GO) on butanol and water by molecular simulations. The selectivity of butanol/water in ILs/GO materials was improved by 4 times compared to that in GO. He *et al.*[54] used quantum chemical methods to study the interaction between four imidazole-based ILs and GO. They found that the anion type, van der Waals interaction and the position of ILs on GO perform the key roles in affecting ILs-GO interactions.

Nevertheless, the simulation studies on the system of ILs/GO and CO_2 are rare, and the correlation between the microstructure of ILs/GO and the CO_2 separation performance, which is crucial information for reasonable designing the new ILs/GO membranes for separating CO_2, is still indistinct. Therefore, in this work, MD simulations were carried out to explore the effects of [Bmim][PF_6], [Bmim][TF_2N] and [Bmim][BF_4] confined in GO on CO_2/CH_4 separation performance. [Bmim][PF_6], [Bmim][BF_4] and [Bmim][TF_2N] were chosen because they are typical ILs with high CO_2 solubility and low CH_4 solubility. The microstructure, interaction and dynamic properties of the ILs and GO were investigated in depth by analyzing the number density, the angular distribution of cations near the interface, the radial distribution function, the hydrogen bond interaction, the interaction energy and the self-diffusion coefficient. Then, the interaction of CO_2/CH_4 with [Bmim][PF_6]/GO, [Bmim][TF_2N]/GO and [Bmim][BF_4]/GO and the diffusion of CO_2/CH_4 in the ILs/GO membranes were further examined. In the following, the simulation details and results are presented at first, and a summary and outlook are given at last.

2. Models and simulation details

The all-atom force fields have been successfully utilized for simulations of GO and ILs. Therefore, the all-atom force field was also adopted in this work. The force field parameters for $[Bmim]^+$, $[PF_6]^-$ and $[BF_4]^-$ based on the AMBER model were obtained from the work of Liu *et al.*[55], and those for GO and $[TF_2N]^-$ were obtained from Fu *et al.*[56] and Lopes *et al.*[57], respectively. These force field parameters have been successfully applied to various systems[58,59]. The chemical structures of $[Bmim]^+$, $[PF_6]^-$, $[TF_2N]^-$, $[BF_4]^-$ and GO are displayed in Fig. 1. According to the experimental evidence[60] and Lerf-Klinowski model [61], the GO surface normally includes hydroxyl, epoxy, carbonyl groups. In this work, the hydroxyl ($c = n_{OH}/n_C = 10\%$) and epoxy ($c = n_O/n_C = 10\%$) groups were considered as the primary functional groups, which are distributed at random on both sides of the GO surface with the dimensions of 124.63Å × 43.51Å. The ILs were confined between the two parallel GO surfaces in the simulated system. The layer spacings of 2, 3 and 4 nm were investigated, and a system with the layer spacing of 2 nm is given in Fig. 1 as an example. The structures of other layer spacings are shown in Figure S1. The number of ILs in [Bmim][PF_6]/GO systems for 2, 3 and 4 nm are 200, 350 and 500, respectively. The number of ILs in [Bmim][BF_4]/GO and [Bmim][TF_2N]/GO systems for 4 nm are 555 and 350, respectively. These are based on the number of bulk ILs simulated in the same volume of boxes. The experimental data [62,63] and simulated densities are summarized in Table S1. The simulated densities are marginally larger than experimental values, with a deviation of about 1.3%, confirming the applicability of the force field to characterize the behavior of [Bmim][PF_6], [Bmim][BF_4] and [Bmim][TF_2N].

All MD simulations were performed by GROMACS 5.1.4 package [64,65]. ILs were placed randomly in the interlayer channel between the GO surfaces by using Packmol[66]. Periodic boundary conditions (PBCs) were utilized in *xy* directions. The cutoff distance of the van der Waals interactions was chosen as 1.2 nm. The long-range electrostatic interaction force was handled by the Particle mesh Ewald method[67,68] with a cut-off radius of 1.2 nm. The temperature was controlled by a Nose-Hoover thermostat[69] with a coupling constant of 0.2 ps. The Parrinello-Rahman barostat[70] was utilized to control the pressure with a coupling constant of 2.0 ps. Pressure coupling was isotropic at 1 bar in the *x* and *y* directions, while it was 0 bar in the *z*-direction, allowing the system to be active in the *z*-direction to better simulate the membrane. All bonds containing hydrogen atoms were limited by the LINCS algorithm[71].

The initial configuration was first energy minimized with convergence criteria $F_{max} < 100 \mathrm{kJ \cdot mol^{-1} \cdot nm^{-1}}$ to avoid the existence of overlapping structures. Then the NVT ensemble was carried out to equilibrate the system for 100 ps at 298 K. The NPT simulation was performed by annealing to bring the systems to equilibrium quickly. All systems were gradually cooled from 500 K to 298 K in 6 ns, followed by 44 ns equilibrating process, and another 50 ns for collecting data. The equilibrium trajectory was recorded every 1 ps.

To examine the interactions and molecular dynamic properties of CO_2/CH_4 in the ILs/GO system, 25, 45, 65 CO_2/CH_4 were inserted into the ILs/GO systems with different layer spacings. The force field parameters for CO_2 and CH_4 were obtained from Shi *et al.*[72] and Skoulidas *et al.*[73], respectively. Simulations were performed for 100 ns, and the last 50 ns were adopted for the analysis.

3. Results and discussion

3.1. ILs/GO systems

3.1.1. Density distribution

Fig. 2 presents the number density profiles of the confined cations and anions between two surfaces of GO at different layer spacings. According to Fig. 2, the densities of anions and cations next to the GO surface are higher than those near the center and the bulk ILs. For example, the value of the number density of cations next to the surface is 4.67 when H = 2 nm (H represents the height of the layer spacing), while

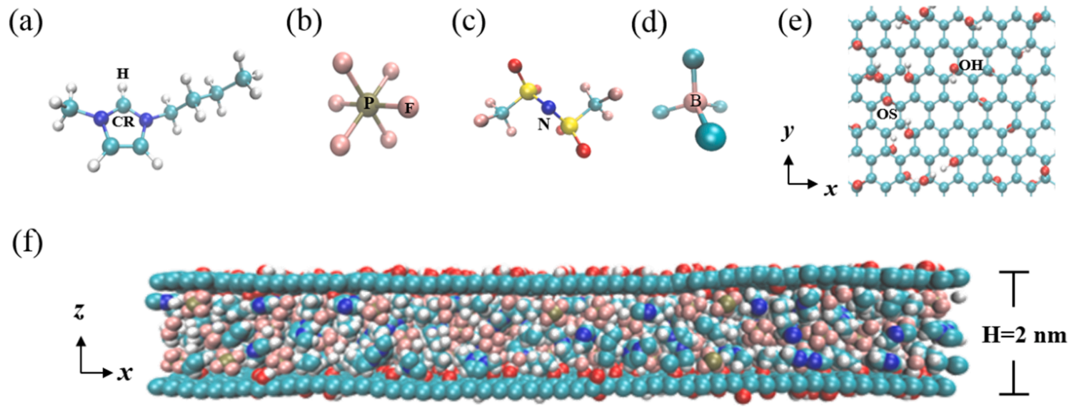

Fig. 1. Structures of cation, anions, GO and the ILs/GO system (a) $[Bmim]^+$, (b) $[PF_6]^-$, (c) $[TF_2N]^-$, (d) $[BF_4]^-$, (e) GO and (f) the confined $[Bmim][PF_6]$/GO system with the layer spacing of 2 nm.

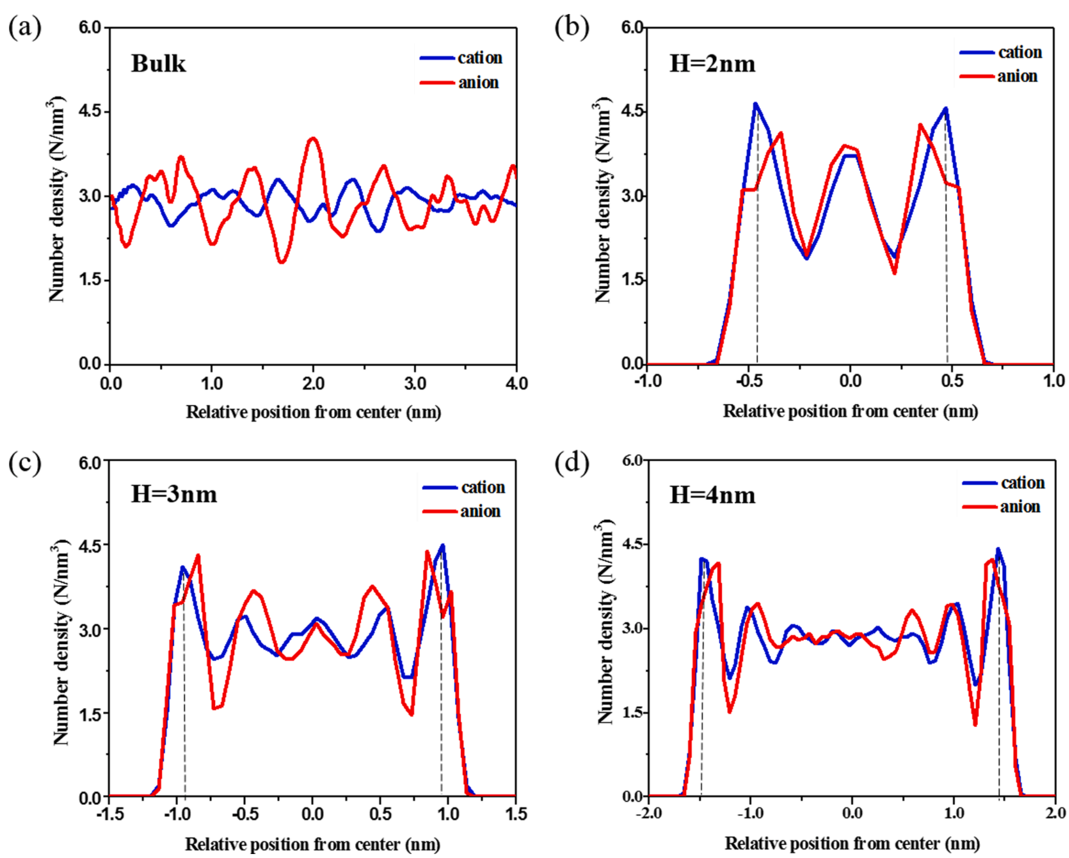

Fig. 2. The number density profiles of cations ($[Bmim]^+$) and anions ($[PF_6]^-$). (a) The bulk ILs, (b)-(d) the confined ILs at different layer spacings (H = 2, 3, 4 nm).

the values at the center and in the bulk ILs are 3.72 and 2.91, respectively.

The presence of the dense ILs adsorption layer at the interface reveals the strong adsorption capacity of GO to ILs. It can also be clearly seen that the density of cations near GO surfaces is higher than that of anions for 2, 3 and 4 nm layer spacings, suggesting that the number of cations is larger than anions within the dense ILs adsorption layer, which may be caused by the electrostatic and π-π interaction between cations and GO [54]. At the same time, a more pronounced oscillation of the curve is present in the smaller interlayer spacing (e.g. at a layer spacing of 2 nm), whereas a more homogeneous density distribution is noted in the larger interlayer spacing and the values of the number density distribution at the center are close to that of the bulk ILs (e.g. at a layer spacing of 4 nm). Furthermore, as the layer spacing increases, the first peaks of the number density distribution for cations and anions appeared almost at the same position, and that for the cations is approximately at 0.55 nm from the GO surfaces. Therefore, two layers of the confined system can be defined with the cation as the object, namely the adsorption layer at 0.55 nm from the GO surfaces and the intermediate bulk phase layer, respectively, which allows further study of the structural properties depending on the distance from the surface. The layered characteristics of anions and cations in confined ILs are consistent with other simulation studies[47,74,75]. According to the FTIR spectra, DSC curves and SSNMR spectra recorded by Peng *et al.* [47], a shift between the peaks of cations and anions close to the GO surfaces was observed due to the nature of the negative electricity on the GO surface. Hence, the simulation results obtained are also agreed with the experimental conclusions of Peng *et al.* [47].

3.1.2. Angular orientation of cations

The presence of the confining surfaces will lead to “angular orientation ordering”. To explore the orientation of the confined cations near the GO surfaces, the orientation vectors and angular schematics together with the orientation of imidazole ring and alkyl side chain of the confined cations were defined and displayed in Fig. 3. Here, three main vectors are considered: (i) the normal vector z of the GO plate (z-axis), (ii) the normal vector of the imidazole ring, (iii) the vector along the alkyl tail, defined by the N atom and the terminal C atom. Meanwhile, we specify: (i) the angle between the normal vector of the imidazole ring and the z-axis as θ. When θ is 0° or 180°, the imidazole ring of cation is parallel to the GO surfaces. When θ is 90°, the imidazole ring of cation is perpendicular to the GO surfaces. (ii) The angle between the vector along the alkyl side chain and the z-axis is φ. When φ is 90°, the alkyl side chain is parallel to the GO surfaces. When φ is 0° or 180°, the alkyl side chain is perpendicular to the GO surfaces.

As illustrated in Fig. 3 (b), the angle between the imidazole ring of cation at the interface and the surface is small at 2 nm, with the angle θ almost close to 0° or 180°. Meanwhile, the angle φ between the alkyl side chain at the interface and the normal vector to the surface is about 90°, as displayed in Fig. 3 (f). The results reveal that the alkyl side chains and imidazole rings are mostly in a state close to parallel to GO surfaces owing to the confinement of the GO surfaces. As the layer spacing increases, the imidazole rings of the cations at the center of the GO channel are substantially randomly distributed, and part of the imidazole rings are perpendicular to the GO surfaces, especially when H = 4 nm while the alkyl side chains still close to parallel to GO. Predictably, the cations of ILs have diverse orientations at different distances from surfaces. These simulations provide further evidence that the cations within the adsorbed layer are arranged more closely and in a regular manner due to the confinement of the GO surfaces. Similar orienting behavior was also observed by experiment[76,77]. For example, Baldelli *et al.* [76] found $[Bmim]^+$ parallel to the graphene surface by SFG vibrational spectroscopy, which was due to π-π interactions of the rings and non-polar interactions of the alkyl chains.

3.1.3. The radial distribution functions

The radial distribution functions (RDFs) usually refer to the distribution probability of other particles in space given the coordinates of a certain particle. For further understanding of the partial structure of the confined ILs, the center-of-mass RDFs for the cations-anions in the ILs/GO system and the bulk ILs were calculated and displayed in Fig. 4. In addition, the site-site RDFs were computed between the O atom in the hydroxyl group on both sides of the GO and the H atom in the cation/F atom in the anion (the position of the H atom is depicted in Fig. 1). As presented in Fig. 4 (a), the first peak of the cations-anions RDFs appears at approximately 0.45 nm and the first trough appears at about 0.75 nm. The position of the first peak of the cations-anions RDFs hardly changed as increasing layer spacing, which may be attributed to the strong electrostatic interaction of anions and cations. Compared with the bulk ILs system, when the ILs are confined between the GO surfaces, the height of the first peak of the cations-anions RDFs increases notably. The smaller the layer spacing, the higher the peak intensity. It describes an increased density of adjacent cations and anions caused by the confined structure, especially when H = 2 nm. With the increase of the layer spacing, the height of the anions-cations RDFs decreases and gradually approaches the bulk ILs, implying the increasingly uniform distribution of ILs. This result further supports the density distribution of the ILs/GO system obtained in Fig. 2 and agrees with the study of other researchers [75,78]. For example, Salemi *et al.* [75] calculated the RDFs of $[Emim]^+$ and $[PF_6]^-$ confined between graphite sheets by MD simulations and found that the height of anions-cations RDFs decreased with increasing layer spacing.

Moreover, for the confined system, the first peak is the highest at 2 nm, while that is the lowest at 4 nm, indicating that the confined ILs with a layer spacing of 4 nm are the weakest aggregated in the studied ILs/GO system. Hence, ILs can be better dispersed in the larger layer spacing, which is conducive to the formation of continuous ILs channels and facilitates the diffusion of CO_2.

In order to survey the interaction of anions and cations with GO, taking the confined system with H = 2 nm as an example, the site-site RDFs between the O atoms(–OH) distributed on both sides of GO and the H atoms in the cation/the F atoms in the anion were investigated. As illustrated in Fig. 4 (b), it is apparent that the first peak of O-H RDFs occurs earlier than that of O-F RDFs, which is because the planar structure of the imidazole cation is closer to the GO than the stereospecific structure of the $[PF_6]^-$. In addition, the first peak of the O-H RDFs is found at 0.21 nm. From the conditions of hydrogen bond formation, it is possible that hydrogen bond was formed between O and H atoms.

3.1.4. Hydrogen bonds interaction

It is common knowledge that there exists strong hydrogen bond (HB) interaction between the anions and cations[13,79], and the HB is believed to notably affect the aggregation structure of imidazolium ILs. To explore the effect of GO surface on the HB interaction between anions and cations, the average number of HB formed between each anion and cation pairs and between each IL pair and the functional groups of the GO surfaces were calculated. Generally, the HB is deemed to form if the distance between acceptor and donor is not greater than 0.35 nm, and the angle of the acceptor-hydrogen-donor is lower than 30°[64]. Here, the potential acceptors are the O atom on the epoxy-functional group of the GO and the F atom of $[PF_6]^-$, the possible donors are the O atom on the hydroxyl functional group of the GO and the CR atom of $[Bmim]^+$ (the position of the CR atom is depicted in Fig. 1). Fig. 5 presents the number of HB formed between each ILs pair and the GO at different layer spacings, as well as the number of HB formed between each anion and cation pair within the ILs before and after confinement. In the bulk ILs system, the average HB formed between each anion and cation pair is 2.64. In the ILs/GO system with layer spacing of 2, 3 and 4 nm, the average HB formed between each anion and cation pair is 2.41, 2.44 and 2.49, respectively. In contrast to that of the bulk ILs, the average number of HB formed between each anion and cation pair in the ILs/GO system is reduced, which resulted in the disruption of the HB network between ILs. Zokaie *et al.* [78] also reported that the reduction in the number of hydrogen bonds between water molecules is caused by confinement in GO. Nevertheless, the number of HB formed between GO and ILs is 0.88, 0.50 and 0.36 with the layer spacing of 2, 3 and 4 nm, respectively. As the layer spacing increases, the number of HB formed between ILs and functional groups of GO decreases. It was further demonstrated that the cations and anions are more evenly distributed in larger layer spacings.

3.1.5. Interaction energy

When the ILs is incorporated between two pieces of GO, it will interact with the GO physically or chemically. Therefore, the interaction energies (coulombic + van der Waals) of the cations-anions, GO-cations and GO-anions were calculated for all systems and shown in Fig. 6. In the ILs/GO system, the GO-cations interaction energy is greater than that of the GO-anions. In addition, the interaction energy between the cations and anions is greater than that between GO and ILs. As the layer spacing increases, the interaction energy of anions-cations increases and the interaction energy of GO-cations/anions decreases. These findings coincide with the results of the number density distribution and the RDF. It also found that the anions-cations interaction energy of the bulk ILs is greater than that of the confined system. Since GO surfaces are negatively charged, the confined $[Bmim]^+$ tends to interact strongly with the GO surfaces via electrostatic interactions, so that the anions-cations interaction energy is reduced. Jiang *et al.* [74] characterized the Ag/ILs-GO interactions within the membrane by Raman and ATR-FTIR,

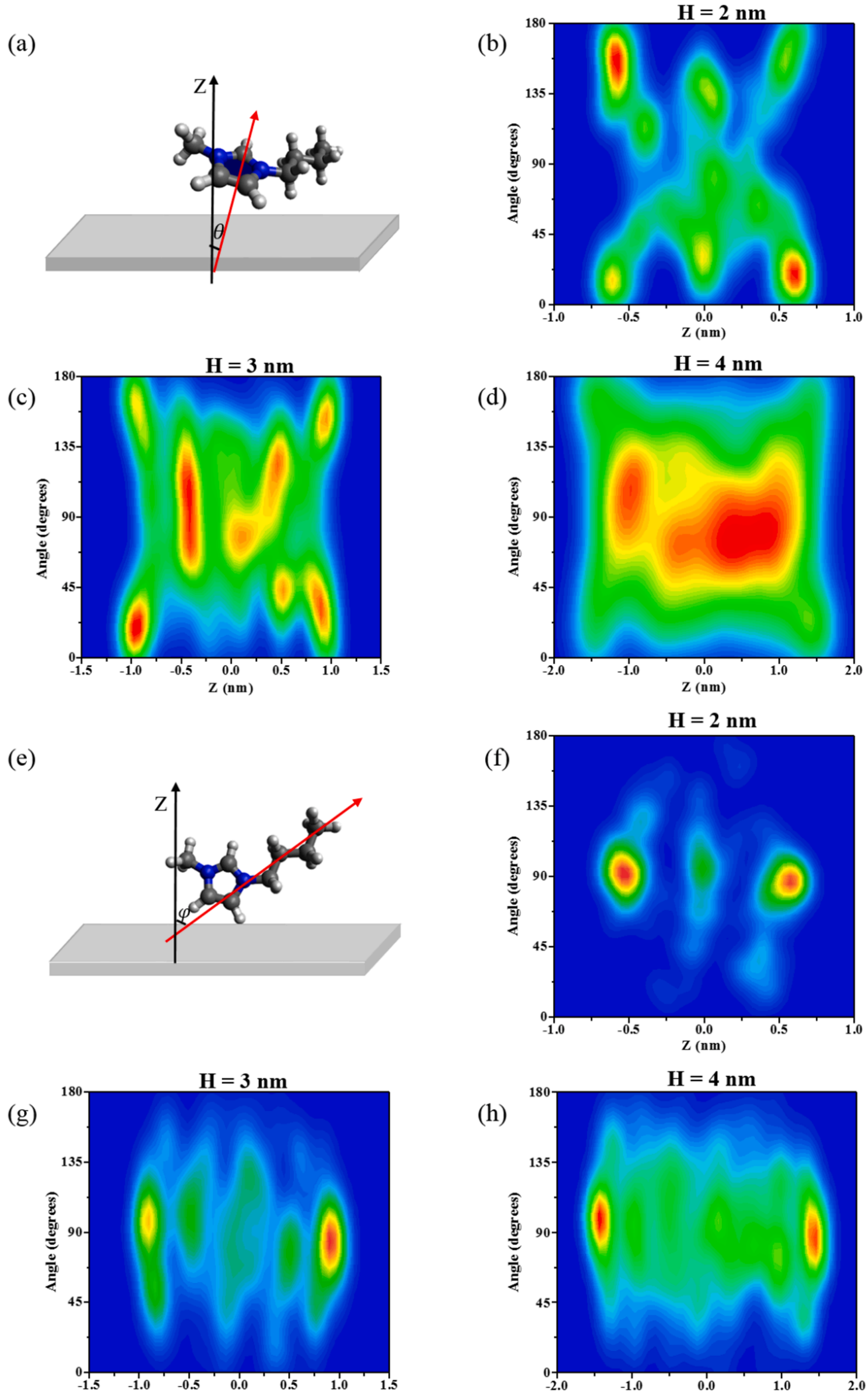

Fig. 3. (a, e) Schematic diagram of the angle of imidazole rings and alkyl side chains of the cations to the GO surface. (b-d, f-h) The angle between the normal vector of the GO plate (z-axis) and the normal vector of the imidazole rings of cations and the vector of the alkyl side chains of the cations when H = 2, 3, and 4 nm. Red = high probability, blue = low probability.

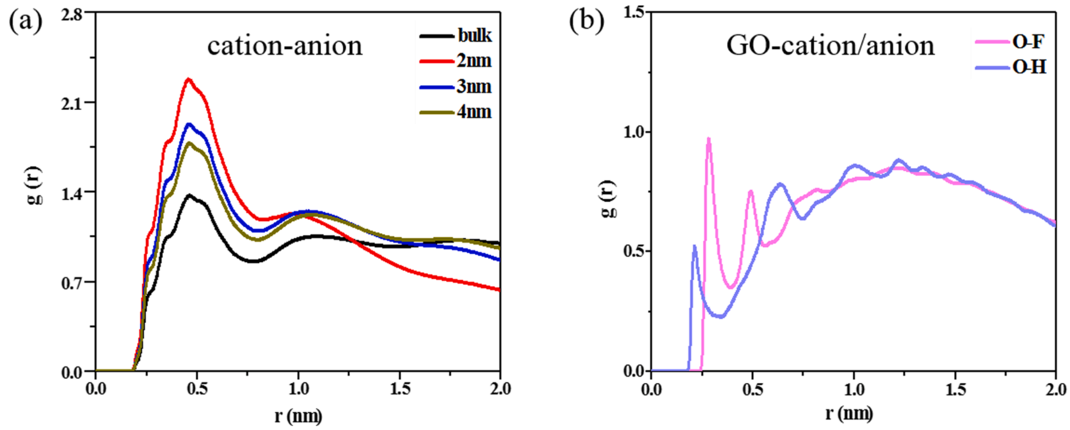

Fig. 4. (a) Center-of-mass RDFs for cations-anions in the bulk ILs and confined system when H = 2, 3, 4 nm, (b) site-site RDFs between the O atom in the hydroxyl group on both sides of the GO and the H atom in the cation/the F atom in the anion when H = 2 nm.

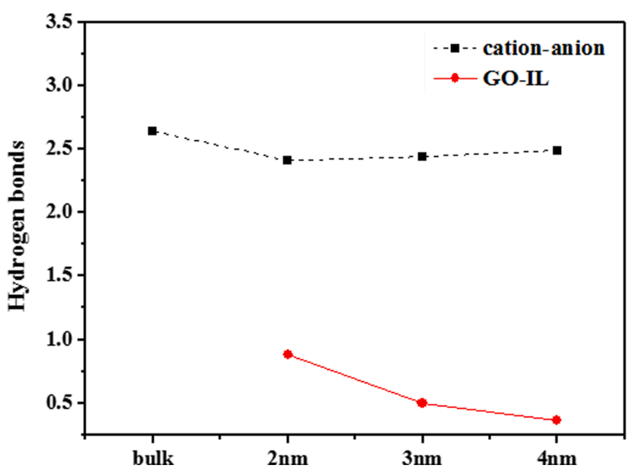

Fig. 5. The number of hydrogen bonds between anions and cations in the bulk ILs and GO/ILs systems with layer spacing of 2, 3, 4 nm, and the number of hydrogen bonds between ILs and functional groups of GO.

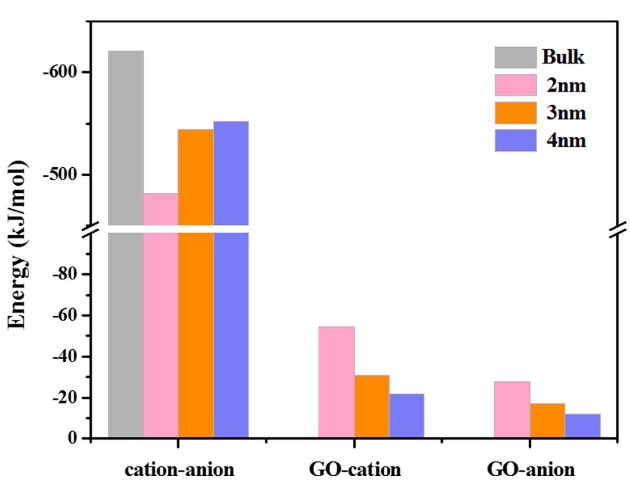

Fig. 6. Interaction energies of cations-anions, GO-cations, GO-anions in the bulk ILs and ILs/GO systems with H = 2, 3 and 4 nm.

and found a blue shift of the anions from 1314 to 1319 cm^{-1}, which was because the interaction of the cations and GO weakens the cations-anions electrostatic interaction of the ILs.

It is known that the anions-cations interaction in ILs affects the free volume[80,81]. The weaker the interaction between anions and cations, the larger the free volume of the ILs, which is more conducive to interacting with CO_2. Therefore, it can be inferred that the confined ILs could increase the solubility of CO_2.

3.1.6. Self-diffusion coefficients

The fluidity of ILs influences significantly the diffusion of the gas. To further evaluate the separation performance of the ILs/GO membrane, the self-diffusion coefficients (D) of anions and cations in all systems were calculated. The self-diffusion coefficients were derived by calculating the mean square displacement (MSD) slope according to the Einstein relationship[82]:

$$D = \frac{1}{6N}\lim_{t\to\infty}\frac{d}{dt}\langle\sum_{i=1}^{N}[r_i(t) - r_i(0)]^2\rangle$$

where N is the total number of molecules, and $r_i(0)$ and $r_i(t)$ are the original and final positions, and the amount in the angle brackets is the tethered average of the MSDs. Several studies have pointed out that to obtain dependable self-diffusion coefficients using Einstein's equation, simulations of long duration are needed to keep the MSD curves in a linear state. Hence, to ensure the dependability and accuracy of the simulated data, a total of 100 ns were simulated, the latter 50 ns were used to calculate the MSD of each component in the system and 10% – 90% of the valid data were intercepted for fitting. As demonstrated in Figure S3, the MSDs curves for all simulated systems are linear in the range of 50 to 100 ns, which indicates that the acquired self-diffusion coefficients for anions and cations are dependable.

Table 1 shows the self-diffusion coefficients of anions and cations in all systems. The diffusion rate of cations is significantly faster than that of anions, which is attributed to the planar structure of $[Bmim]^+$ and the greater translational motion friction of $[PF_6]^-$. The result is in line with the work of other researchers[75,78]. It is also found that the diffusion of both anions and cations in the bulk ILs is significantly faster than those in the ILs/GO system, which is because that the interaction between the cations/anions and GO reduces the mobility of the anions and cations. Moreover, as the layer spacing of the ILs/GO system increases,

Table 1
Self-Diffusion Coefficients (1×10^{-13} m^2/s) of Cations and Anions for [Bmim][PF_6]/GO Systems.

System	D[cation]	D[anion]
Bulk	11.88 ± 1.28	3.95 ± 0.33
2 nm	0.65 ± 0.01	0.17 ± 0.02
3 nm	0.95 ± 0.01	0.35 ± 0.01
4 nm	1.53 ± 0.01	0.43 ± 0.01

the diffusion rate of both anions and cations increases, which is following the results in Fig. 4 (a). The same phenomenon was found for [Bmim][TF_2N]/GO and [Bmim][BF_4]/GO, and the results are listed in Table S2.

3.2. ILs/GO-CO_2/CH_4 systems

3.2.1. Number density distribution of CO_2/CH_4

The number density distributions of CO_2/CH_4 were calculated to learn about the distribution of CO_2/CH_4 in the ILs/GO system. The number density distribution curves of CO_2 and CH_4 at a layer spacing of 2 nm are illustrated in Fig. 7. The number density distribution curves for other layer spacings are presented in Figure S4. As depicted in Fig. 7, the center-of-mass distribution of CO_2/CH_4 coincides with the density peaks of cations and anions, indicating that CO_2/CH_4 is mainly transported and diffused through the ILs channel of the ILs/GO membrane. This finding was also observed in the ILs/GO system with layer spacing of 3 and 4 nm, which is also consistent with the work of Peng *et al.* [47], who found that the ILs channel in GO-SILM provides high selectivity and rapid permeation by experiment.

3.2.2. Interactions of CO_2/CH_4 with confined ILs

To further study the interaction of CO_2/CH_4 within the confined ILs, the site-site RDFs of CO_2-cations, CO_2-anions, CH_4-cations and CH_4-anions were analyzed and displayed in Fig. 8. Here, the C(CO_2), C(CH_4), P and CR represent the CO_2, CH_4, $[PF_6]^-$ and $[Bmim]^+$ sites, respectively. The peak profiles of the RDFs for the gas and the bulk ILs/the confined ILs are similar, except for the height of the peaks. As depicted in Fig. 8 (a, b), the heights of the RDFs peaks for the CO_2/CH_4 and the cations/anions of the bulk ILs are significantly lower than those of the confined ILs. The results demonstrate that the interaction between CO_2/CH_4 and the confined ILs is stronger than that with the bulk ILs, implying that the confined ILs/GO system is more beneficial for capturing gases. In addition, as illustrated in Fig. 8 (b-d), as the layer spacing increases, the first peak values for CO_2/CH_4-cations and CO_2/CH_4-anions decreases, implying the interaction between gas and confined ILs decreases. This conclusion is corroborated by the calculated anions-CO_2 interaction energies for bulk ILs and confined ILs with different layer spacings shown in Figure S7. Therefore, the smaller the layer spacing of the ILs/GO membrane, the higher the solubility for CO_2/CH_4. Moreover, it is obvious that the first peak values for CO_2-anions and CO_2-cations are higher than those for CH_4-anions and CH_4-cations, suggesting that the interaction between CO_2 and confined ILs is much stronger than that of CH_4. Thus, the ILs/GO membrane could form the channel for CO_2 transport with a high CO_2 solubility, which is in accordance with the findings of Peng *et al*[47]. They confirmed experimentally by sorption and adsorption isotherms and time lag methods that impregnated ILs sealed large defects and folds in GO membranes, resulting in a significant increase in CO_2 solubility, which was approximately twice that of bulk ILs[47].

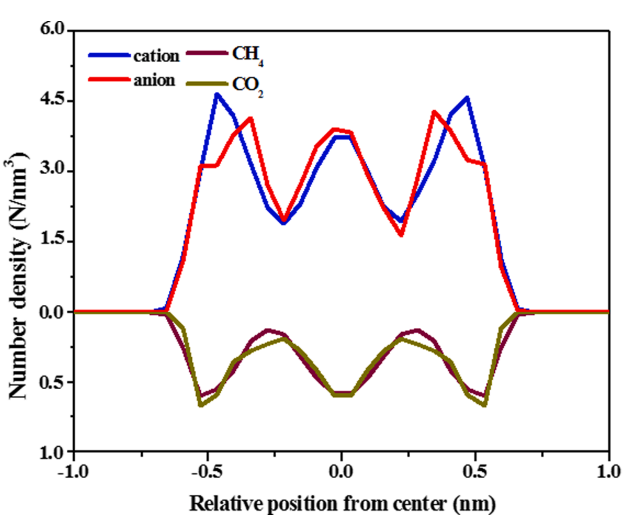

Fig. 7. The number density profiles of cations ($[Bmim]^+$), anions ($[PF_6]^-$), CO_2 and CH_4 at 2 nm.

In addition, the relationship between the solubility and the layer spacing may depend on the GO-IL combination. As shown in Fig. 6, the GO-cations/anions interaction energy decreases with increasing layer spacing, but the cations-anions interaction increases. It is well known that the weaker the interaction between anions and cations, the larger the free volume of ILs, which is more favorable for capturing more CO_2. Therefore, the smaller the layer spacing of the ILs/GO membranes, the higher the solubility for CO_2/CH_4.

3.2.3. Dynamic properties of CO_2/CH_4

The kinetic properties of the gas in the ILs/GO systems are important because the slow diffusion can be a bottleneck in the adsorption process. To understand CO_2/CH_4 transport inside the ILs/GO systems, the self-diffusion coefficients (D) of CO_2/CH_4 in all systems were calculated and compared with that in the bulk ILs. As demonstrated in Figure S5, the MSDs curves for all simulated systems are linear. As shown in Table 2, the diffusion rate of CO_2/CH_4 in ILs/GO systems is less than that of bulk ILs, which is probably because of the stronger interaction of CO_2/CH_4-confined ILs and the reduced diffusion of ILs in confined GO. Furthermore, the diffusion rate of CO_2/CH_4 increase as the GO layer spacing increases, which is because that as the layer spacing increases, the diffusion rate of ILs increases, thus promoting the diffusion of CO_2/CH_4. In addition, the diffusion rate of CO_2 is slightly larger than that of CH_4 in all simulated systems. Therefore, it can be summarized that the ILs channels in ILs/GO system contribute more to the diffusion of CO_2, reflecting the diffusion selectivity of the ILs/GO membrane for CO_2. The results are consistent with the experimental findings that the confinement reduces the self-diffusivity of the gas but does not reduce selectivity[47].

Based on the above discussion, it is found that the interaction energies of CO_2 with anions are significantly higher than those of CH_4, while the difference in diffusion rate for CO_2 and CH_4 is not substantial, which indicates that solubility plays a dominant role and diffusion plays a secondary role. Peng et al.[47] also found in their experiments that CO_2/CH_4 selectivity is mainly determined by the solubility of the gas. In our simulated system, the strongest interaction between CO_2 and anions is found at 2 nm. Therefore, 2 nm is considered to be the optimal layer spacing.

3.2.4. Effect of anion types on CO_2/CH_4 separation in ILs/GO membrane

It is well known that the anions play a dominant role in CO_2 absorption for bulk ILs, thus the effect of ILs/GO membranes with different types of anions on CO_2/CH_4 separation was also investigated. Site-site RDFs and the interaction energies for CO_2/CH_4-anions were calculated for [Bmim][PF_6]/GO, [Bmim][TF_2N]/GO and [Bmim][BF_4]/GO systems with a layer spacing of 4 nm. The results were presented in Fig. 9. The C(CO_2), C(CH_4), P, N and B represent the CO_2, CH_4, $[PF_6]^-$, $[TF_2N]^-$ and $[BF_4]^-$ sites, respectively. As shown in Fig. 9 (a-b), the highest RDFs peak positions between C(CO_2) and P, N and B atoms are at 0.41, 0.51 and 0.39 nm, respectively, while those for C(CH_4) are at 0.47, 0.63 and 0.46 nm, respectively. In addition, the height of the RDFs peaks for CH_4-anions is significantly lower than that for CO_2-anions. The results suggest the interaction of CO_2 with the anions is stronger than those of CH_4. Moreover, from Fig. 9 (c-d) it can be found that the interaction energies of CO_2 with $[PF_6]^-$, $[TF_2N]^-$ and $[BF_4]^-$ are greater than those of CH_4, and especially for CO_2-$[TF_2N]^-$. The order of the interaction energies of CO_2 with anions is $[TF_2N]^- > [PF_6]^- > [BF_4]^-$, which reveals that the [Bmim][TF_2N]/GO membrane performs better solubility selectivity for CO_2/CH_4 than [Bmim][PF_6]/GO and [Bmim][BF_4]/GO membranes.

The low diffusion rate of gases in confined membrane materials is

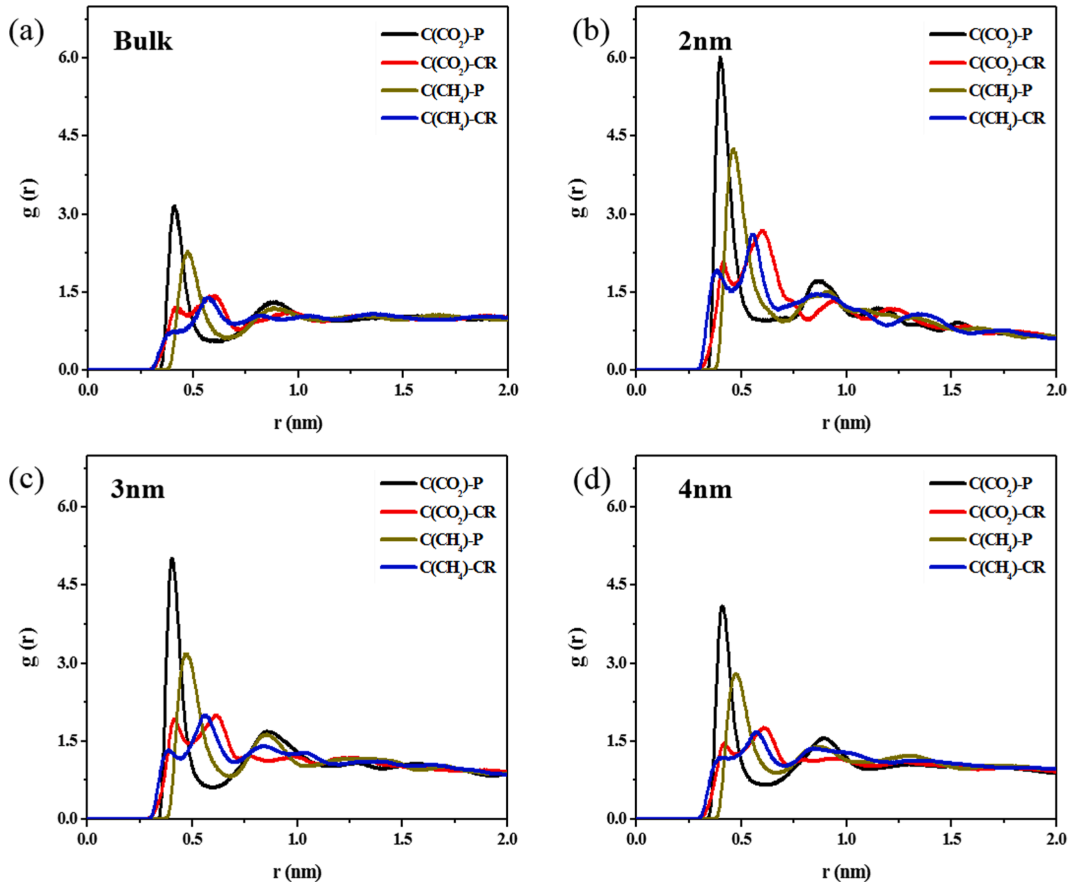

Fig. 8. Site-site RDFs between the C atom in the CO_2/CH_4 and the CR atom in the cation/the P atom in the anion. (a) The bulk ILs, (b-d) the confined ILs at different layer spacings (H = 2, 3, 4 nm).

Table 2
CO_2/CH_4 Self-Diffusion Coefficients (1×10^{-11} m^2/s) in All [Bmim][PF_6]/GO Systems.

System	D[CO_2]	D[CH_4]
Bulk	2.10 ± 0.02	1.35 ± 0.01
2 nm	0.09 ± 0.02	0.04 ± 0.00
3 nm	0.37 ± 0.01	0.23 ± 0.01
4 nm	0.95 ± 0.01	0.83 ± 0.01

always a challenge for fabricating separation membranes, thus the selection of suitable ILs is particularly important. The self-diffusion coefficients of CO_2/CH_4 in [Bmim][PF_6]/GO, [Bmim][TF_2N]/GO and [Bmim][BF_4]/GO systems were also calculated and listed in Table 3. As demonstrated in Figure S6, the MSDs curves for all simulated systems are linear, indicating that the calculated self-diffusion coefficients are reliable. In addition, the simulated self-diffusion coefficients were compared with experiment results. For example, the self-diffusion coefficients of CO_2 and CH_4 in [Bmim][BF_4]/GO at 2 nm are 0.20×10^{-11} m^2/s and 0.11×10^{-11} m^2/s at 298 K and 0.1 MPa. Peng et al.[47] obtained self-diffusion coefficients by time-lag experiment, and reported that self-diffusion coefficients of CO_2 and CH_4 in [Bmim][BF_4]/GO at 1.94 nm are 0.23×10^{-10} m^2/s and 0.79×10^{-11} m^2/s at 298 K and 0.5 MPa. Due to the difference in pressure conditions, the simulation results can be considered to be consistent with the experimental data, which further suggests the simulated self-diffusion coefficients are reliable. From Table 3, it can be found that the diffusion of CO_2 is always faster than CH_4 in three systems and CO_2 diffuses fastest in the [Bmim][TF_2N]/GO membrane, which reveals that the [Bmim][TF_2N]/GO membrane exhibits better diffusion selectivity for CO_2/CH_4. Therefore, compared to [Bmim][PF_6] and [Bmim][BF_4], it can be predicted that the [Bmim][TF_2N]/GO membrane provides superior solubility and diffusion selectivity for CO_2/CH_4 separation.

In addition, it is well known that the viscosity of ILs mainly affects the gas diffusion, so a lower viscosity ionic liquid 1-butyl-3-methylimidazolium tricyandiamide ([Bmim][TCM]) was added to calculate the self-diffusion coefficient of CO_2/CH_4 in the ILs/GO systems. The order of viscosity is [Bmim][PF_6] (213 mPa·s) > [Bmim][BF_4] (112 mPa·s)> [Bmim][TF_2N] (45 mPa·s)>[Bmim][TCM] (27.84 mPa·s)[83]. As shown in Table 2 and 3, the order of CO_2/CH_4 self-diffusion coefficient is [Bmim][PF_6]/GO < [Bmim][BF_4]/GO < [Bmim][TF_2N]/GO < [Bmim][TCM]/GO, which suggest that the lower the viscosity of ILs, the faster the CO_2/CH_4 diffusion. Therefore, the low viscosity ILs were found to be beneficial to improve the diffusion selectivity of CO_2/CH_4.

4. Conclusions

The systems of CO_2/CH_4 in three ILs/GO membranes with different layer spacings were studied by all-atom MD simulations. The layer spacings were 2, 3 and 4 nm, respectively. The number density distributions of anions and cations confined in GO were investigated. It was found that there were dense cation adsorption layers near the GO surface, and the imidazole rings and alkyl side chains were mostly paralleled to the GO surface. The analysis of the cations-anions RDFs of the bulk ILs and the ILs/GO system with three different layer spacings showed that the position of the first peak of cations-anions RDFs hardly changed with increasing layer spacing, while the height decreased and gradually approached bulk ILs, implying that ILs became more uniform. In addition, the average number of hydrogen bonds formed between each anion/cation pair in the ILs/GO system was reduced compared to

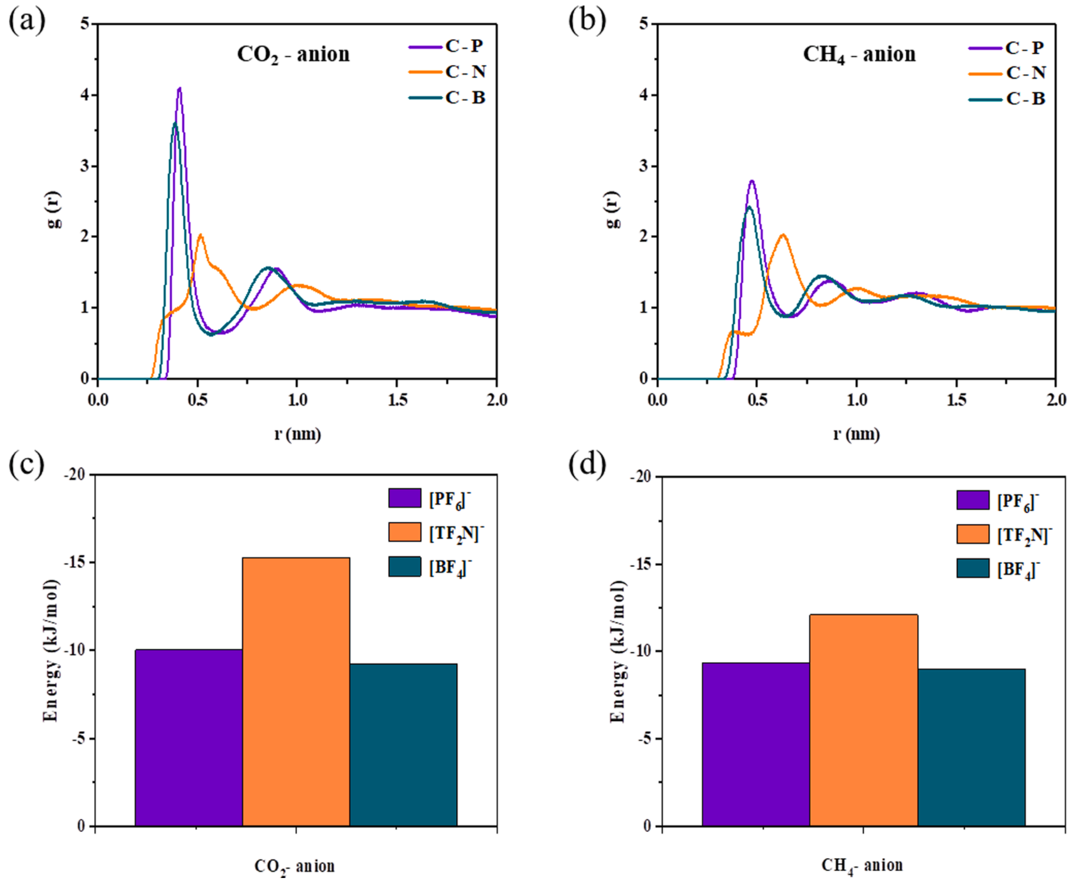

Fig. 9. (a-b) Site-site RDFs between the C atom in the CO_2/CH_4 and the P/N/B atom in $[PF_6]^-$, $[TF_2N]^-$ and $[BF_4]^-$, (c-d) the interaction energies of CO_2/CH_4 with $[PF_6]^-$, $[TF_2N]^-$ and $[BF_4]^-$.

Table 3
CO_2/CH_4 Self-Diffusion Coefficients (1×10^{-11} m^2/s) in ILs/GO Membranes.

System	[Bmim][BF_4]/GO		[Bmim][TF_2N]/GO		[Bmim][TCM]/GO	
	D[CO_2]	D[CH_4]	D[CO_2]	D[CH_4]	D[CO_2]	D[CH_4]
2 nm	0.20 ± 0.03	0.11 ± 0.06	0.20 ± 0.04	0.12 ± 0.04	0.29 ± 0.15	0.19 ± 0.03
3 nm	0.51 ± 0.04	0.26 ± 0.00	1.28 ± 0.09	0.79 ± 0.06	4.71 ± 0.46	3.79 ± 0.96
4 nm	1.24 ± 0.01	0.89 ± 0.08	2.61 ± 0.06	2.41 ± 0.01	10.12 ± 0.27	4.59 ± 0.20

that of bulk ILs. Whereas, it was found that hydrogen bond was formed between ILs and GO, and the number of hydrogen bonds between ILs and GO decreased with increasing layer spacing, which further demonstrates that ILs can better disperse in membrane. The results of interaction energy suggested that the interaction of the cations with GO is greater than that of the anions with GO, which leads to the lower anions-cations interaction energy of the ILs/GO system, thus increasing CO_2 solubility. Moreover, as the layer spacing of the ILs/GO system increases, the diffusion of both cations and anions becomes faster, which facilitates the diffusion of CO_2.

Furthermore, the dynamic properties and interaction of CO_2/CH_4 in ILs/GO system were also examined. It was found that the interaction of CO_2/CH_4 with the confined ILs was stronger than that with the bulk ILs, which indicates that the confined ILs could enhance the gas solubility. In addition, the ILs-CO_2 interaction is stronger than ILs-CH_4 interaction, and the ILs-CO_2 interaction increased with decreasing layer spacing, indicating that the ILs/GO membrane with smaller layer spacing has higher solubility selectivity for CO_2/CH_4. Meanwhile, the self-diffusion coefficients of CO_2/CH_4 in the ILs/GO system were discovered to be smaller than that of bulk ILs, and the diffusion of CO_2 is faster than that of CH_4 in all systems. However, the diffusion of CO_2/CH_4 became faster as the layer spacing increased, which reflects with larger layer spacing would result in the higher diffusion selectivity for CO_2/CH_4. Whereas, the solubility selectivity plays a dominant role, thus the optimal layer spacing is 2 nm. Furthermore, the low viscosity ILs were found to be beneficial to improve the diffusion selectivity. Moreover, the solubility selectivity and diffusion selectivity of CO_2/CH_4 in [Bmim][TF_2N]/GO membrane are higher than those in [Bmim][PF_6]/GO and [Bmim][BF_4]/GO membranes, which can be predicted that [Bmim][TF_2N]/GO membrane has better performance in separating CO_2/CH_4. Therefore, the low viscosity [Bmim][TF_2N]/GO membrane with 2 nm is recommended for separating CO_2/CH_4 for practical applications. These findings present a molecular perspective on the ILs-GO interaction, and also provide theoretical support for preparation of the ILs/GO membranes with super CO_2/CH_4 separation performance.

CRediT authorship contribution statement

Fang Yan: Investigation, Writing – original draft. **Yandong Guo:** Supervision, Writing – review & editing. **Zhenlei Wang:** Investigation, Validation. **Linlin Zhao:** Investigation, Validation. **Xiaochun Zhang:** Supervision, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Natural Science Foundation of China (21978293, 22078024), Innovation Academy for Green Manufacture (IAGM2020C18), the LiaoNing Revitalization Talents Program (XLYC2007175), the National Natural Science Foundation of Liaoning Province of China (2021MS316), and the Educational Commission of Liaoning Province of China (LQ2020001). The authors sincerely appreciate Prof. Suojiang Zhang (IPE, CAS) for his careful academic guidance and great support.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.seppur.2022.120736.

References

[1] X. Chen, G. Liu, W. Jin, Natural gas purification by asymmetric membranes: an overview, Green Energy Environ. 6 (2021) 176–192.
[2] X.-B. Zhang, J. Xu, Optimal policies for climate change: a joint consideration of CO_2 and methane, Appl. Energy 211 (2018) 1021–1029.
[3] Y. Li, S. Lan, M. Ryberg, J. Perez-Ramirez, X. Wang, A quantitative roadmap for China towards carbon neutrality in 2060 using methanol and ammonia as energy carriers, iScience 24 (2021) 102513.
[4] Y. Xie, Z. Hou, H. Liu, C. Cao, J. Qi, The sustainability assessment of CO_2 capture, utilization and storage (CCUS) and the conversion of cropland to forestland program (CCFP) in the water–energy–food (WEF) framework towards China's carbon neutrality by 2060, Environ. Earth Sci. 80 (2021).
[5] M. Batres, F.M. Wang, H. Buck, R. Kapila, U. Kosar, R. Licker, D. Nagabhushan, E. Rekhelman, V. Suarez, Environmental and climate justice and technological carbon removal, Electr. J. 34 (2021).
[6] L. Deng, H. Kvamsdal, CO_2 capture: challenges and opportunities, Green Energy Environ. 1 (2016).
[7] A. Sayari, Y. Belmabkhout, R. Serna-Guerrero, Flue gas treatment via CO_2 adsorption, Chem. Eng. J. 171 (2011) 760–774.
[8] A.A. Olajire, CO_2 capture and separation technologies for end-of-pipe applications – a review, Energy 35 (2010) 2610–2628.
[9] S. Zeng, X. Zhang, L. Bai, X. Zhang, H. Wang, J. Wang, D. Bao, M. Li, X. Liu, S. Zhang, Ionic-liquid-based CO_2 capture systems: structure, interaction and process, Chem. Rev. 117 (2017) 9625–9673.
[10] R. Khalilpour, K. Mumford, H. Zhai, A. Abbas, G. Stevens, E.S. Rubin, Membrane-based carbon capture from flue gas: a review, J. Cleaner Prod. 103 (2015) 286–300.
[11] M. Chawla, H. Saulat, M. Masood Khan, M. Mahmood Khan, S. Rafiq, L. Cheng, T. Iqbal, M.I. Rasheed, M.Z. Farooq, M. Saeed, N.M. Ahmad, M.B. Khan Niazi, S. Saqib, F. Jamil, A. Mukhtar, N. Muhammad, Membranes for CO_2/CH_4 and CO_2/N_2 gas separation, Chem. Eng. Technol. 43 (2019) 184–199.
[12] L. You, Y. Guo, Y. He, F. Huo, S. Zeng, C. Li, X. Zhang, X. Zhang, Molecular level understanding of CO_2 capture in ionic liquid/polyimide composite membrane, Front. Chem. Sci. Eng. (2021).
[13] T. Song, X. Zhang, Y. Li, K. Jiang, S. Zhang, X. Cui, L. Bai, Separation efficiency of CO_2 in ionic liquids/poly(vinylidene fluoride) composite membrane: a molecular dynamics study, Ind. Eng. Chem. Res. 58 (2019) 6887–6898.
[14] Y. Cheng, Y. Guo, H. He, W. Ding, Y. Diao, F. Huo, Mechanistic understanding of CO_2 adsorption and diffusion in the imidazole ionic liquid-hexafluoroisopropylidene polyimide composite membrane, Ind. Eng. Chem. Res. 60 (2021) 6027–6037.
[15] S. Saqib, S. Rafiq, M. Chawla, M. Saeed, N. Muhammad, S. Khurram, K. Majeed, A. L. Khan, M. Ghauri, F. Jamil, M. Aslam, Facile CO_2 separation in composite membranes, Chem. Eng. Technol. 42 (2019) 30–44.
[16] Y. Han, W.S.W. Ho, Polymeric membranes for CO_2 separation and capture, J. Membr. Sci. 628 (2021).
[17] S. Kim, C.A. Scholes, D.E. Heath, S.E. Kentish, Gas-liquid membrane contactors for carbon dioxide separation: a review, Chem. Eng. J. 411 (2021).
[18] P.H. Presumido, A. Primo, V.J.P. Vilar, H. Garcia, Large area continuous multilayer graphene membrane for water desalination, Chem. Eng. J. 413 (2021).
[19] W. Guo, S.M. Mahurin, R.R. Unocic, H. Luo, S. Dai, Broadening the gas separation utility of monolayer nanoporous graphene membranes by an ionic liquid gating, Nano Lett. 20 (2020) 7995–8000.
[20] W. Ying, X. Peng, Graphene oxide nanoslit-confined $AgBF_4$/ionic liquid for efficiently separating olefin from paraffin, Nanotechnology 31 (2019) 085703.
[21] W. Fam, J. Mansouri, H. Li, J. Hou, V. Chen, Gelled graphene oxide-ionic liquid composite membranes with enriched ionic liquid surfaces for improved CO_2 separation, ACS Appl. Mater. Interfaces 10 (2018) 7389–7400.
[22] H. Liu, S. Dai, D.E. Jiang, Insights into CO_2/N_2 separation through nanoporous graphene from molecular dynamics, Nanoscale 5 (2013) 9984–9987.
[23] L. Dai, K. Huang, Y. Xia, Z. Xu, Two-dimensional material separation membranes for renewable energy purification, storage, and conversion, Green Energy Environ. 6 (2021) 193–211.
[24] F. Moghadam, H.B. Park, Two-dimensional materials: an emerging platform for gas separation membranes, Curr. Opin. Chem. Eng. 20 (2018) 28–38.
[25] S. Xu, W.C. Li, C.T. Wang, L. Tang, G.P. Hao, A.H. Lu, Self-Pillared Ultramicroporous Carbon Nanoplates for Selective Separation of CH_4/N_2, Angewandte Chemie (International ed. in English), 60 (2021) 6339-6343.
[26] Z. Gan, Y. Wang, M. Wang, E. Gao, F. Huo, W. Ding, H. He, S. Zhang, Ionophobic nanopores enhancing the capacitance and charging dynamics in supercapacitors with ionic liquids, J. Mater. Chem. A 9 (2021) 15985–15992.
[27] C. Chen, J. Wang, D. Liu, C. Yang, Y. Liu, R.S. Ruoff, W. Lei, Functionalized boron nitride membranes with ultrafast solvent transport performance for molecular separation, Nat. Commun. 9 (2018) 1902.
[28] A.W. Ameen, J. Ji, M. Tamaddondar, S. Moshenpour, A.B. Foster, X. Fan, P. M. Budd, D. Mattia, P. Gorgojo, 2D boron nitride nanosheets in PIM-1 membranes for CO_2/CH_4 separation, J. Membr. Sci. 636 (2021).
[29] S. Bi, H. Banda, M. Chen, L. Niu, M. Chen, T. Wu, J. Wang, R. Wang, J. Feng, T. Chen, M. Dinca, A.A. Kornyshev, G. Feng, Molecular understanding of charge storage and charging dynamics in supercapacitors with MOF electrodes and ionic liquid electrolytes, Nat. Mater. 19 (2020) 552–558.
[30] J.H. Qin, H. Zhang, P. Sun, Y.D. Huang, Q. Shen, X.G. Yang, L.F. Ma, Ionic liquid induced highly dense assembly of porphyrin in MOF nanosheets for photodynamic therapy, Dalton Trans 49 (2020) 17772–17778.
[31] M.H. Yap, K.L. Fow, G.Z. Chen, Synthesis and applications of MOF-derived porous nanostructures, Green Energy Environ. 2 (2017) 218–245.
[32] Y. Ban, Z. Li, Y. Li, Y. Peng, H. Jin, W. Jiao, A. Guo, P. Wang, Q. Yang, C. Zhong, W. Yang, Confinement of Ionic Liquids in Nanocages: Tailoring the Molecular Sieving Properties of ZIF-8 for Membrane-Based CO_2 Capture, Angewandte Chemie (International ed. in English), 54 (2015) 15483-15487.
[33] T. Yu, Q. Cai, G. Lian, Y. Bai, X. Zhang, X. Zhang, L. Liu, S. Zhang, Mechanisms behind high CO_2/CH_4 selectivity using ZIF-8 metal organic frameworks with encapsulated ionic liquids: a computational study, Chem. Eng. J. 419 (2021).
[34] Y. Liu, Y. Zhao, X. Zhang, X. Huang, W. Liao, Y. Zhao, MoS_2-based membranes in water treatment and purification, Chem. Eng. J. 422 (2021).
[35] N. Aguilar, S. Aparicio, Theoretical insights into CO_2 adsorption by MoS_2 nanomaterials, J. Phys. Chem. C 123 (2019) 26338–26350.
[36] K. Yin, S. Huang, X. Chen, X. Wang, J. Kong, Y. Chen, J. Xue, Generating sub-nanometer pores in single-layer MoS_2 by heavy-ion bombardment for gas separation: a theoretical perspective, ACS Appl. Mater. Interfaces 10 (2018) 28909–28917.
[37] D. Chen, W. Ying, Y. Guo, Y. Ying, X. Peng, Enhanced gas separation through nanoconfined ionic liquid in laminated MoS_2 membrane, ACS Appl. Mater. Interfaces 9 (2017) 44251–44257.
[38] X. Yan, S. Anguille, M. Bendahan, P. Moulin, Ionic liquids combined with membrane separation processes: a review, Sep. Purif. Technol. 222 (2019) 230–253.
[39] J. Tong, Y. Zhao, F. Huo, Y. Guo, X. Liang, N. von Solms, H. He, The dynamic behavior and intrinsic mechanism of CO_2 absorption by amino acid ionic liquids, Phys. Chem. Chem. Phys.: PCCP 23 (2021) 3246–3255.
[40] X. Zhang, X. Liu, X. Yao, S. Zhang, Microscopic structure, interaction, and properties of a guanidinium-based ionic liquid and its mixture with CO_2, Ind. Eng. Chem. Res. 50 (2011) 8323–8332.
[41] H. Gao, L. Bai, J. Han, B. Yang, S. Zhang, X. Zhang, Functionalized ionic liquid membranes for CO_2 separation, Chem. Commun. (Camb.) 54 (2018) 12671–12685.
[42] X. Zhang, K. Jiang, Z. Liu, X. Yao, X. Liu, S. Zeng, K. Dong, S. Zhang, Insight into the performance of acid gas in ionic liquids by molecular simulation, Ind. Eng. Chem. Res. 58 (2018) 1443–1453.
[43] L.A. Blanchard, D. Hancu, E.J. Beckman, J.F. Brennecke, Green processing using ionic liquids and CO_2, Nature 399 (1999) 28–29.
[44] J.L. Anthony, J.L. Anderson, E.J. Maginn, J.F. Brennecke, Anion effects on gas solubility in ionic liquids, J. Phys. Chem. B 109 (2005) 6366–6374.
[45] I.S. Chae, J.H. Lee, J. Hong, Y.S. Kang, S.W. Kang, The platform effect of graphene oxide on CO_2 transport on copper nanocomposites in ionic liquids, Chem. Eng. J. 251 (2014) 343–347.
[46] W. Ying, K. Zhou, Q. Hou, D. Chen, Y. Guo, J. Zhang, Y. Yan, Z. Xu, X. Peng, Selectively tuning gas transport through ionic liquid filled graphene oxide nanoslits using an electric field, J. Mater. Chem. A 7 (2019) 15062–15067.
[47] W. Ying, J. Cai, K. Zhou, D. Chen, Y. Ying, Y. Guo, X. Kong, Z. Xu, X. Peng, Ionic liquid selectively facilitates CO_2 transport through graphene oxide membrane, ACS Nano 12 (2018) 5385–5393.
[48] Y.-Y. Lee, B. Gurkan, Graphene oxide reinforced facilitated transport membrane with poly(ionic liquid) and ionic liquid carriers for CO_2/N_2 separation, J. Membr. Sci. 638 (2021).
[49] K. Dong, S. Zhang, J. Wang, Understanding the hydrogen bonds in ionic liquids and their roles in properties and reactions, Chem. Commun. (Camb.) 52 (2016) 6744–6764.
[50] X. Zhang, Z. Liu, X. Liu, Understanding the interactions between tris (pentafluoroethyl)-trifluorophosphate-based ionic liquid and small molecules from molecular dynamics simulation, Sci. China Chem. 55 (2012) 1557–1565.
[51] Y. Wang, F. Huo, H. He, S. Zhang, The confined [Bmim][BF_4] ionic liquid flow through graphene oxide nanochannels: a molecular dynamics study, Phys. Chem. Chem. Phys. 20 (2018) 17773–17780.
[52] Y. Wang, C. Wang, Y. Zhang, F. Huo, H. He, S. Zhang, Molecular insights into the regulatable interfacial property and flow behavior of confined ionic liquids in graphene nanochannels, Small 15 (2019) e1804508.
[53] W. Tang, H. Lou, Y. Li, X. Kong, Y. Wu, X. Gu, Ionic liquid modified graphene oxide-PEBA mixed matrix membrane for pervaporation of butanol aqueous solutions, J. Membr. Sci. 581 (2019) 93–104.

[54] Y. He, Y. Guo, F. Yan, T. Yu, L. Liu, X. Zhang, T. Zheng, Density functional theory study of adsorption of ionic liquids on graphene oxide surface, Chem. Eng. Sci. 245 (2021).
[55] Z.P. Liu, S.P. Huang, W.C. Wang, A refined force field for molecular simulation of imidazolium-based ionic liquids, J. Phys. Chem. B 108 (2004) 12978–12989.
[56] Z. Fu, W. Xu, G. Chen, Z. Wang, D. Lu, J. Wu, Z. Liu, Molecular dynamics simulations reveal how graphene oxide stabilizes and activates lipase in an anhydrous gas, Phys. Chem. Chem. Phys. 21 (2019) 25425–25430.
[57] J.N.C. Lopes, A.A.H. Padua, Molecular force field for ionic liquids composed of triflate or bistriflylimide anions, J. Phys. Chem. B 108 (2004) 16893–16898.
[58] K. Miyazaki, N. Takenaka, T. Fujie, E. Watanabe, Y. Yamada, A. Yamada, M. Nagaoka, Impact of cis- versus trans-configuration of butylene carbonate electrolyte on microscopic solid electrolyte interphase formation processes in lithium-ion batteries, ACS Appl. Mater. Interfaces 11 (2019) 15623–15629.
[59] A.Z. Haddad, A.K. Menon, H. Kang, J.J. Urban, R.S. Prasher, R. Kostecki, Solar desalination using thermally responsive ionic liquids regenerated with a photonic heater, Environ. Sci. Technol. 55 (2021) 3260–3269.
[60] W. Gao, L.B. Alemany, L. Ci, P.M. Ajayan, New insights into the structure and reduction of graphite oxide, Nat. Chem. 1 (2009) 403–408.
[61] A. Lerf, H.Y. He, M. Forster, J. Klinowski, Structure of graphite oxide revisited, J. Phys. Chem. B 102 (1998) 4477–4482.
[62] Z.Y. Gu, J.F. Brennecke, Volume expansivities and isothermal compressibilities of imidazolium and pyridinium-based ionic liquids, J. Chem. Eng. Data 47 (2002) 339–345.
[63] J.G. Huddleston, A.E. Visser, W.M. Reichert, H.D. Willauer, G.A. Broker, R. D. Rogers, Characterization and comparison of hydrophilic and hydrophobic room temperature ionic liquids incorporating the imidazolium cation, Green Chem. 3 (2001) 156–164.
[64] B. Hess, C. Kutzner, D. van der Spoel, E. Lindahl, GROMACS 4: Algorithms for highly efficient, load-balanced, and scalable molecular simulation, J. Chem. Theory Comput. 4 (2008) 435–447.
[65] D. Van Der Spoel, E. Lindahl, B. Hess, G. Groenhof, A.E. Mark, H.J. Berendsen, GROMACS: fast, flexible, and free, J. Comput. Biophys. Chem. 26 (2005) 1701–1718.
[66] L. Martinez, R. Andrade, E.G. Birgin, J.M. Martinez, PACKMOL: a package for building initial configurations for molecular dynamics simulations, J. Comput. Biophys. Chem. 30 (2009) 2157–2164.
[67] K.J. Oh, Y. Deng, An efficient parallel implementation of the smooth particle mesh Ewald method for molecular dynamics simulations, Comput. Phys. Commun. 177 (2007) 426–431.
[68] U. Essmann, L. Perera, M.L. Berkowitz, T. Darden, H. Lee, L.G. Pedersen, A smooth particle mesh Ewald method, J. Chem. Phys. 103 (1995) 8577–8593.
[69] C. Braga, K.P. Travis, A configurational temperature Nose-Hoover thermostat, J. Chem. Phys. 123 (2005) 134101.
[70] M. Parrinello, A. Rahman, Polymorphic transitions in single crystals: A new molecular dynamics method, J. Appl. Phys. 52 (1981) 7182–7190.
[71] B. Hess, H. Bekker, H.J.C. Berendsen, J.G.E.M. Fraaije, LINCS: A linear constraint solver for molecular simulations, J. Comput. Chem. 18 (1997) 1463–1472.
[72] W. Shi, E.J. Maginn, Atomistic simulation of the absorption of carbon dioxide and water in the ionic liquid 1-n-hexyl-3-methylimidazolium bis (trifluoromethylsulfonyl)imide (hmim Tf2N), J. Phys. Chem. B 112 (2008) 2045–2055.
[73] A.I. Skoulidas, D.S. Sholl, Transport diffusivities of CH_4, CF_4, He, Ne, Ar, Xe, and SF_6 in silicalite from atomistic simulations, J. Phys. Chem. B 106 (2002) 5058–5067.
[74] H. Dou, M. Xu, B. Jiang, G. Wen, L. Zhao, B. Wang, A. Yu, Z. Bai, Y. Sun, L. Zhang, Z. Chen, Z. Jiang, Bioinspired graphene oxide membranes with dual transport mechanisms for precise molecular separation, Adv. Funct. Mater. 29 (2019) 1905229.
[75] S. Salemi, H. Akbarzadeh, S. Abdollahzadeh, Nano-confined ionic liquid [emim][PF_6] between graphite sheets: a molecular dynamics study, J. Mol. Liq. 215 (2016) 512–519.
[76] S. Baldelli, J. Bao, W. Wu, S.-S. Pei, Sum frequency generation study on the orientation of room-temperature ionic liquid at the graphene–ionic liquid interface, Chem. Phys. Lett. 516 (2011) 171–173.
[77] R. Atkin, G.G. Warr, Structure in confined room-temperature ionic liquids, J. Phys. Chem. C 111 (2007) 5162–5168.
[78] M. Zokaie, M. Foroutan, Comparative study on confinement effects of graphene and graphene oxide on structure and dynamics of water, RSC Adv. 5 (2015) 39330–39341.
[79] Z. Dai, Y. You, Y. Zhu, S. Wang, W. Zhu, X. Lu, Atomistic insights into the layered microstructure and time-dependent stability of [BMIM][PF_6] confined within the meso-slit of carbon, J. Phys. Chem. B 123 (2019) 6857–6869.
[80] S. Zeng, J. Wang, L. Bai, B. Wang, H. Gao, D. Shang, X. Zhang, S. Zhang, Highly selective capture of CO_2 by ether-functionalized pyridinium ionic liquids with low viscosity, Energy Fuels 29 (2015) 6039–6048.
[81] X.H. Huang, C.J. Margulis, Y.H. Li, B.J. Berne, Why is the partial molar volume of CO_2 so small when dissolved in a room temperature ionic liquid? structure and dynamics of CO_2 dissolved in $[Bmim]^+[PF_6]^-$, J. Am. Chem. Soc. 127 (2005) 17842–17851.
[82] T.I. Morrow, E.J. Maginn, Molecular dynamics study of the ionic liquid 1-n-butyl-3-methylimidazolium hexafluorophosphate, J. Phys. Chem. B 106 (2002) 12807–12813.
[83] L.F. Zubeir, G.E. Romanos, W.M.A. Weggemans, B. Iliev, T.J.S. Schubert, M. C. Kroon, Solubility and diffusivity of CO_2 in the ionic liquid 1-Butyl-3-methylimidazolium tricyanomethanide within a large pressure range (0.01 MPa to 10 MPa), J. Chem. Eng. Data 60 (2015) 1544–1562.